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Lab : 3.4.1.6

Python INSTITUTE

3.4.1.6 LAB: The basics of lists

MODULE (59%)

SECTION (43%)

Estimated time

5 minutes

Level of difficulty

Very easy

Objectives

Familiarize the student with:

- using basic instructions related to lists;
- creating and modifying lists.

Scenario

There once was a hat. The hat contained no rabbit, but a list of five numbers: `[1, 2, 3, 4]`, and `5`.

Your task is to:

- write a line of code that prompts the user to replace the middle number in the list with an integer number entered by the user (Step 1)
- write a line of code that removes the last element from the list (Step 2)
- write a line of code that prints the length of the existing list (Step 3).

Ready for this challenge?

```
1 hat_list = [1, 2, 3, 4, 5] # This is an existing list of numbers hidden in the hat.
2
3 # Step 1: write a line of code that prompts the user
4 # to replace the middle number with an integer number entered by the user.
5
6 replace_num = int(input("Enter the number that you want to replace the middle number with: "))
7 hat_list[2] = replace_num
8 # Step 2: write a line of code that removes the last element from the list.
9 del hat_list[-1]
10 # Step 3: write a line of code that prints the length of the existing list.
11 print(f"Length of list: {len(hat_list)}")
12 print(hat_list)
13
```

Console>_

Enter the number that you want to replace the middle number with: 111
Length of list: 4
[1, 2, 111, 4]

Lab : 3.4.1.13

Python INSTITUTE

3.4.1.13 LAB: The basics of lists - the Beatles

MODULE (68%)

SECTION (93%)

creating and modifying simple lists;

using methods to modify lists.

Scenario

The Beatles were one of the most popular music group of the 1960s, and the best-selling band in history. Some people consider them to be the most influential act of the rock era. Indeed, they were included in *Time* magazine's compilation of the 20th Century's 100 most influential people.

The band underwent many line-up changes, culminating in 1962 with the line-up of John Lennon, Paul McCartney, George Harrison, and Richard Starkey (better known as Ringo Starr).

Write a program that reflects these changes and lets you practice with the concept of lists. Your task is to:

- step 1: create an empty list named `beatles`;
- step 2: use the `append()` method to add the following members of the band to the list: `John Lennon`, `Paul McCartney`, and `George Harrison`;
- step 3: use the `for` loop and the `append()` method to prompt the user to add the following members of the band to the list: `Stu Sutcliffe`, and `Pete Best`;
- step 4: use the `del` instruction to remove `Stu Sutcliffe` and `Pete Best` from the list;
- step 5: use the `insert()` method to add `Ringo Starr` to the beginning of the list.

By the way, are you a Beatles fan? (The Beatles is one of Greg's favorite bands. But wait...who's Greg...?)

```
1 # step 1
2 beatles = []
3 print("Step 1:", beatles)
4
5 # step 2
6 beatles.append("John Lennon")
7 beatles.append("Paul McCartney")
8 beatles.append("George Harrison")
9 print("Step 2:", beatles)
10
11 # step 3
12 for i in range(2):
13     if i==0:
14         member = input("Enter Stu Sutcliffe to add him in the list: ")
15     else:
16         member = input("Enter Pete Best to add him in the list: ")
17     beatles.append(member)
18
19 print("Step 3:", beatles)
20
21 # step 4
22 for _ in range(2):
23     del beatles[len(beatles)-1]
24
25 print("Step 4:", beatles)
26
27 # step 5
28 beatles.insert(0, "Ringo Starr")
29 print("Step 5:", beatles)
30
```

Console>_

Enter Stu Sutcliffe to add him in the list: Stu Sutcliffe
Enter Pete Best to add him in the list: Pete Best
Step 3: ['John Lennon', 'Paul McCartney', 'George Harrison', 'Stu Sutcliffe', 'Pete Best']
Step 4: ['John Lennon', 'Paul McCartney', 'George Harrison']
Step 5: ['Ringo Starr', 'John Lennon', 'Paul McCartney', 'George Harrison']
The Fab 4

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3.4.1.13 LAB: The basics of lists - the Beatles >

MODULE (48%)SECTION (93%)

- creating and modifying simple lists;
- using methods to modify lists.

Scenario

The Beatles were one of the most popular music group of the 1960s, and the best-selling band in history. Some people consider them to be the most influential act of the rock era. Indeed, they were included in *Time* magazine's compilation of the 20th Century's 100 most influential people.

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Write a program that reflects these changes and lets you practice with the concept of lists. Your task is to:

- step 1: create an empty list named `beatles`;
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- step 5: use the `insert()` method to add `Ringo Starr` to the beginning of the list.

By the way, are you a Beatles fan? (The Beatles is one of Greg's favorite bands. But wait...who's Greg...?)

```
1 # step 2
2 beatles.append("John Lennon")
3 beatles.append("Paul McCartney")
4 beatles.append("George Harrison")
5 print("Step 2:", beatles)
6
7 # step 3
8 for i in range(2):
9     if i==0:
10         member = input("Enter Stu Sutcliffe to add him in the list: ")
11     else:
12         member = input("Enter Pete Best to add him in the list: ")
13     beatles.append(member)
14
15 print("Step 3:", beatles)
16
17 # step 4
18 for i in range(2):
19     del beatles[len(beatles)-1]
20
21 print("Step 4:", beatles)
22
23 # step 5
24 beatles.insert(0, "Ringo Starr")
25 print("Step 5:", beatles)
26
27 # testing list length
28 print("The Fab", len(beatles))
29
30
31
32
33
34
```

Console>_

Enter Stu Sutcliffe to add him in the list: Stu Sutcliffe

Enter Pete Best to add him in the list: Pete Best

Step 3: ['John Lennon', 'Paul McCartney', 'George Harrison', 'Stu Sutcliffe', 'Pete Best']

Step 4: ['John Lennon', 'Paul McCartney', 'George Harrison']

Step 5: ['Ringo Starr', 'John Lennon', 'Paul McCartney', 'George Harrison']

The Fab 4

1 # step 2

2 member = input("Enter Stu Sutcliffe to add him in the list: ")

Console>_

Step 1: []

Step 2: ['John Lennon', 'Paul McCartney', 'George Harrison']

Enter Stu Sutcliffe to add him in the list: Stu Sutcliffe

Enter Pete Best to add him in the list: Pete Best

Step 3: ['John Lennon', 'Paul McCartney', 'George Harrison', 'Stu Sutcliffe', 'Pete Best']

Step 4: ['John Lennon', 'Paul McCartney', 'George Harrison']

Step 5: ['Ringo Starr', 'John Lennon', 'Paul McCartney', 'George Harrison']

The Fab 4